

Unit 1

INTRODUCTION TO DBMS

1. Introduction to Data Processing

Context

- Nowadays, more and more data is collected for **academic, scientific research, private & personal use, institutional use, and commercial use**.
- This collected data needs to be **stored, sorted, filtered, analysed and presented** for it to be of any use.
- This process can be simple or complex depending on the **scale** at which data collection is done and the **complexity** of the results required.

Output Files

Different types of output files are obtained from data:

- Plain text file / HTML file
- Table / Spreadsheet
- Charts & Graphs

Definition of Data Processing

- Data processing is the **conversion of data into usable and desired form**.
- This conversion or “processing” is carried out using a predefined sequence of operations either **manually or automatically**.

Stages of Data Processing

The complete process can be divided into **6 basic simple stages**:

1. **Data collection**
2. **Storage of data**
3. **Sorting of data**
4. **Processing of data**
5. **Data analysis**
6. **Data presentation and conclusions**

Types of Data Processing

1. **Manual data processing:** Data is processed manually without the use of machines or electronic devices. This method might be accompanied by automatic methods for completion.
 2. **Mechanical data processing:** Processing is done by use of mechanical devices or very simple electronic devices like calculators and typewriters. This is adopted when the need for processing is simple.
 3. **Electronic data processing (EDP):** This is the **fastest and best available method** with the highest reliability and accuracy. It uses the latest technology (computers/software) and is employed in most agencies.
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2. File Systems vs. DBMS

Overview of File Systems

- A typical file-processing system is supported by a **conventional operating system**.
- The system stores permanent records in various files, and needs different application programs to extract records from and add records to the appropriate files.
- Before DBMSs were introduced, organizations usually stored information in such systems.

Disadvantages of File Systems

The file system approach has several major disadvantages:

1. **Data redundancy and inconsistency:** The same data is duplicated in different files.
2. **Difficulty in accessing data:** No standard way to retrieve specific data easily.
3. **Data isolation:**
 - *Definition:* Isolation is the **database-level property** that controls how and when changes are made, and if they become visible to each other, users, and systems.
 - *Goal:* To allow multiple transactions to occur at the same time without adversely affecting the execution of each.
4. **Integrity problems:**
 - *Definition:* Data integrity is the overall **accuracy, completeness, and consistency** of data.
5. **Concurrent-access anomalies:** Issues arise when multiple users access data simultaneously.
6. **Security problems:** Hard to enforce access controls.

Purpose of DBMS

There are several advantages of a Database Management System over a file system:

- **No redundant data:** Redundancy is removed by **data normalization**.
- **Data Consistency and Integrity:** Data normalization takes care of this too.
- **Secure:** Each user has a different set of access.
- **Privacy:** Limited access.

- Easy access to data.
 - Easy recovery.
 - Flexible.
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3. Basic Database Concepts

Key Terminology

- **Field (Column):**
 - A **single piece of information** (e.g., a name or a number).
 - In some cases, it may even be a **null or empty value**.
- **Record (Row):**
 - A **collection of related fields**.
 - A number of pieces of information that relate to the **same object**.
 - *Example:* An employee's name, address, SSN, and phone number together form the employee's record.
- **Table (File):**
 - A **collection of related records**.
 - *Example:* If you put all employee records together, you have a table of employees.

Database vs. Relational Database

- **Database:** A collection of tables.
 - *Example:* Tables for employees, customers, and sales records combined form a database.
 - **Relational Database:** A collection of related tables.
 - The difference lies in **how the tables are constructed**.
 - If you keep records on a series of Excel spreadsheets, you have just a **database** (you cannot use info from one sheet to ask questions about another).
 - In a **Relational Database**, tables have a **logical link** between them. You can follow this link to get relevant information from another table.
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4. DBMS Features and Architecture

Features of a DBMS

- Creating and Adding of Data Structures.
- Query language.
- Reporting facility.
- Security, Control, Backup, and Disaster Recovery.

Data Abstraction (3 Levels)

A Database Management System (DBMS) uses **three levels of abstraction** to separate how data is stored from how it is seen by users and applications. This hides complexity and supports *data independence*.

1. External Level (View Level)

- This is the **highest level** of abstraction.
- Describes **only the part of the database that a specific user or application needs to see**.
- Each user or application can have a *custom view* of the data.
- Users don't see how data is stored or how other parts of the database are organized.

Example: A sales clerk sees product name and price; a manager sees total sales figures.

2. Conceptual Level (Logical Level)

- This is the **middle level** of abstraction.
- Describes **what data is stored** in the entire database and **how different pieces of data are logically related**.
- It defines entities (tables), attributes (columns), and relationships.
- It hides physical storage details from users.

Example: A table of employees with fields like `Emp_ID`, `Emp_Name`, `Manager_ID` and relationships between departments.

3. Internal Level (Physical Level)

- This is the **lowest level** of abstraction.
- Describes **how data is physically stored** on disk.
- Defines storage structures like files, indices (for example, B+ trees and hashing), and access methods used by the DBMS.
- Users and application programs do not see these details.

Example: How records are organized in blocks, how indexes speed up search, how data files are allocated on disk.

Level	What It Shows
External Level	User-specific views of the data
Conceptual Level	Logical structure of the whole database
Internal Level	Physical storage and data structures

Data Models

A collection of tools for describing Data, Data relationships, and Data constraints.

- **Relational model.**
 - **Entity-Relationship (ER) data model** (mainly for database design).
 - **Object-based data models** (Object-oriented and Object-relational).
 - **Semi structured data model** (XML).
 - **Other older models:** Network model and Hierarchical model.
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5. Database Languages

A DBMS has languages to read, store, and update data.

1. Data Definition Language (DDL)

Definition:

Used to define database structure or pattern.

Usage:

Used to create the skeleton of the database (schema, tables, indexes, constraints).

Metadata:

Stores information like the number of tables, schema names, columns, etc.

Commands and Syntax:

CREATE

Create a table

```
CREATE TABLE table_name (  
    column_name datatype,  
    ...  
);
```

ALTER

Alter the table structure

```
ALTER TABLE table_name  
ADD column_name datatype;
```

DROP

Drop the table

```
DROP TABLE table_name;
```

TRUNCATE

Drop all rows of a table

```
TRUNCATE TABLE table_name;
```

RENAME

Rename the table

```
RENAME table_name TO new_table_name;
```

COMMENT

Add a description to the table

```
COMMENT ON TABLE table_name  
IS 'comment text';
```

2. Data Manipulation Language (DML)

Definition:

Used for accessing and manipulating data. Handles user requests.

Commands and Syntax:

SELECT

Retrieve data.

```
SELECT column1, column2  
FROM table_name  
WHERE condition;
```

INSERT

Insert data into a table.

```
INSERT INTO table_name (column1, column2)  
VALUES (value1, value2);
```

UPDATE

Update existing data.

```
UPDATE table_name  
SET column1 = value1  
WHERE condition;
```

DELETE

Delete all records from a table for a condition.

```
DELETE FROM table_name  
WHERE condition;
```

MERGE

Performs INSERT or UPDATE operations.

```
MERGE INTO target_table t
USING source_table s
ON (condition)
WHEN MATCHED THEN
    UPDATE SET t.column = s.column
WHEN NOT MATCHED THEN
    INSERT (column)
    VALUES (s.column);
```

3. Data Control Language (DCL)

Definition:

Used to retrieve stored/saved data and control access.

Note on Rollback:

DCL execution is transactional and has rollback parameters. However, in Oracle database, DCL execution does not have the feature of rolling back.

Commands and Syntax:

GRANT

Give user access privileges.

```
GRANT privilege
ON object_name
TO user_name;
```

REVOKE

Take back permissions from the user.

```
REVOKE privilege
ON object_name
FROM user_name;
```

4. Transaction Control Language (TCL)

Definition:

Used to run the changes made by DML statements. Grouped into a logical transaction.

Commands and Syntax:

COMMIT

Save the transaction to the database.

```
COMMIT;
```

ROLLBACK

Restore the database to original since the last Commit.

```
ROLLBACK;
```

5 A) Datatypes

Database Data Types

Category	Data Type	Description	Example Syntax
Numeric	INT / INTEGER	Stores whole numbers	INT
Numeric	SMALLINT	Stores smaller range of integers	SMALLINT
Numeric	BIGINT	Stores very large integers	BIGINT
Numeric	DECIMAL(p, s) / NUMERIC(p, s)	Fixed-point numeric values	DECIMAL(10, 2)
Numeric	FLOAT	Approximate floating-point numbers	FLOAT
Numeric	REAL / DOUBLE	Double-precision floating-point numbers	DOUBLE
Character	CHAR(n)	Fixed-length character string	CHAR(10)
Character	VARCHAR(n)	Variable-length character string	VARCHAR(255)
Character	TEXT	Large text data	TEXT
Date & Time	DATE	Stores date (YYYY-MM-DD)	DATE
Date & Time	TIME	Stores time (HH:MI:SS)	TIME
Date & Time	TIMESTAMP	Stores date and time	TIMESTAMP
Date & Time	DATETIME	Date and time combined (DB-specific)	DATETIME
Boolean	BOOLEAN / BOOL	Stores TRUE or FALSE	BOOLEAN
Binary	BINARY(n)	Fixed-length binary data	BINARY(16)
Binary	VARBINARY(n)	Variable-length binary data	VARBINARY(255)
Binary	BLOB	Binary Large Object (media files)	BLOB

Category	Data Type	Description	Example Syntax
Special	ENUM	Single value from predefined list	<code>ENUM('A','B','C')</code>
Special	SET	Multiple values from predefined list	<code>SET('A','B','C')</code>
Special	JSON	Stores JSON documents	<code>JSON</code>
Special	UUID	Stores universally unique identifier	<code>UUID</code>
Special	XML	Stores XML data	<code>XML</code>

6. Database Users and Administrators

Database Users

People who work with a database are categorized as:

1. **Naive Users:** Unsophisticated users who interact by invoking previously written **application programs** (e.g., a clerk in a university).
2. **Application Programmers:** Computer professionals who write application programs and choose tools to develop user interfaces.
3. **Sophisticated Users:** Interact **without** writing programs. They use a **query language** or data analysis software. Includes analysts who explore data.
4. **Specialized Users:** Sophisticated users who write specialized applications that do not fit the traditional framework (e.g., **CAD systems**, **knowledge base/expert systems**, systems with **complex data types**).

Database Administrator (DBA)

The DBA allows the system to regulate which parts of the database various users can access.

Functions of a DBA:

1. **Schema definition:** Creates the original database schema using **DDL**.
 2. **Storage structure and access-method definition.**
 3. **Schema and physical-organization modification:** Changes structure to reflect changing needs or improve **performance**.
 4. **Granting of authorization:** Regulates user access.
 5. **Routine maintenance:**
 - **Backing up** the database (tapes or remote servers) to prevent loss in disasters like **flooding**.
 - Ensuring enough **free disk space** and upgrading as required.
 - **Monitoring jobs** to ensure performance is not degraded by expensive tasks.
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7. Applications and Challenges

Database Applications

DBMS contains information about a particular enterprise. It is a collection of interrelated data and a set of programs to access it.

- **Banking:** Transactions.
- **Airlines:** Reservations, schedules.
- **Universities:** Registration, grades.
- **Sales:** Customers, products, purchases.
- **Online retailers:** Order tracking, customized recommendations.
- **Manufacturing:** Production, inventory, orders, supply chain.
- **Human resources:** Employee records, salaries, tax deductions.

Challenges in Building a DBMS

- Growing complexity in landscape.
- Limits on scalability.
- Increasing data volumes.
- Data security.
- Decentralized data management.